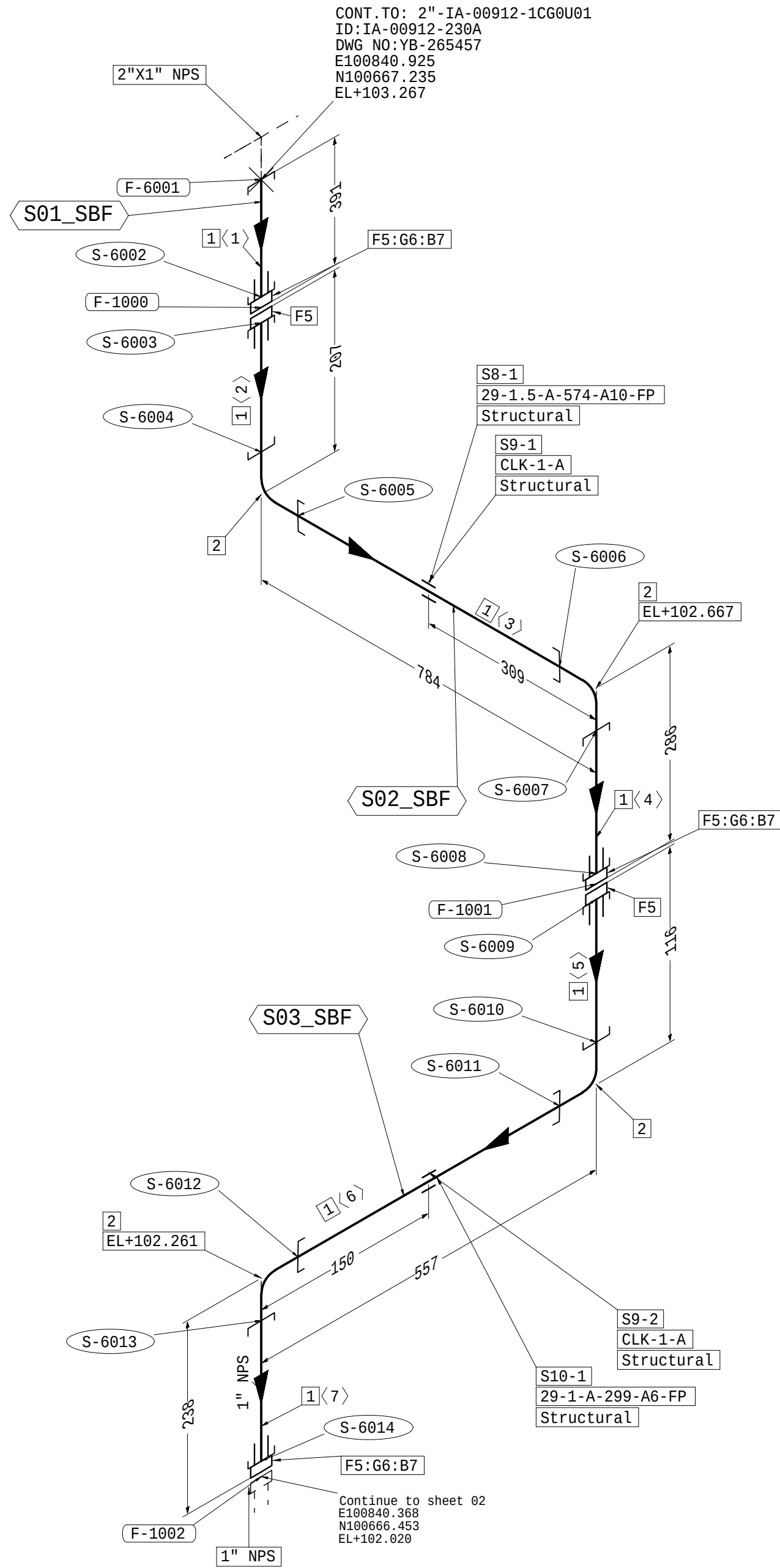
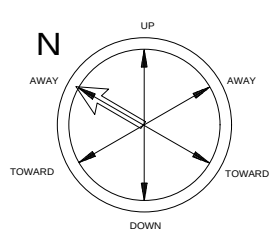


THIS SPOOL MARKUP DRAWING IS GENERATED FROM ENGINEERING PDF & REFERRING APPROVED ENGG ISOMETRICS. ENGINEERING ISOMETRICS TO BE REFERRED FOR REVISION APPROVAL INFORMATION & IN CASE OF ANY DISCREPANCY DATA GIVEN IN ENGINEERING ISOMETRIC SHALL GOVERN



All Pipe attached supports shall be welded before galvanising.

NOTES CONT. :

91. FOR SOUR SERVICE PWHT IS MANDATORY FOR 1. DOUBLE SIDED WELDING, 2. INTERNAL WELD REPAIR 3. INTERNAL WELD BUILD UP FOR DISSIMILAR THICKNESS JOINTS TO OVERCOME MISALIGNMENT.

PWHT	CHEMICAL CLEANING	PIPING MATERIAL CLASS	DESIGN		OPERATING		TEST PRESS. (PSIG)	TEST MEDIUM	RADIOGRAPHY %	INSULATION		TRACING	PAINTING	STRESS SYS NO.	LINE ID
			TEMP. (DEG. F)	PRESS. (PSIG)	TEMP. (DEG.F)	PRESS. (PSIG)				SHOP	FIELD	THK. (MM)	TYPE		
N	NOTE-8	1CG0U01	185	145	135	125	125	SERVICE	NA	NA	0	NI	NA	APCS-26	IA-00970-230A

MATERIALS

PT No.	DESCRIPTION	NPD(IN)	ITEM CODE	QTY
PIPES				
1	PIPE SMLS 1" A106-B GALV STD NPT (M) B36.10M	1	PPS2AEAE01ZZABUUA	2.364
FITTINGS				
2	ELB 90° FRG LR 1" A105N GALV 3000# NPT (F) B16.11	1	PE72ADAE01ZZAA0ATA	4
FLANGES				
5	FLG SCRD RF 1" A105N GALV 150# 125 B16.5	1	PFH2ADAE01ZZAAADH	5
GASKET				
6	GSK SPR WND W/CS CENTERING RING + I - SS 1" 4.50 MM THICK SS316L+GRPH FILL 150# B16.20	1	PGUUB5AG01AIAAAZZC	3
FASTSNERS				
7	BOLT, STUD+2 NUTS A193 B7/ A194 2H, CERAMIC FLP COAT B18.2.1/18.2.2 - 80 mm Length	0.5	PH8AB5AGAE00080ZZF	12
SUPPORTS				
8	PIPE SUPPORT	1	29-1.5-A-574-A10-FP	1
9	PIPE SUPPORT	1	CLK-1-A	2
10	PIPE SUPPORT	1	29-1-A-299-A6-FP	1

PIPE CUT LENGTH SUMMARY						SXX_Spool sequence Number (S01,02, SXX_SP - Fab. Spool	
SPL No.	NPD	PC No.	LENGTH	ALLOWANCE	TOTAL LENGTH	SXX_SP - Fab. Spool	
S01_SBF	1	<1>	0.391	0.000	0.391	SXX_SP - Fab. Spool	
S02_SBF	1	<4>	0.259	0.000	0.259	SXX_SP - Fab. Spool	
S02_SBF	1	<3>	0.730	0.000	0.730	SXX_SP - Fab. Spool	
S02_SBF	1	<2>	0.180	0.000	0.180	SXX_SP - Fab. Spool	
S03_SBF	1	<7>	0.211	0.000	0.211	SXX_SP - Fab. Spool	
S03_SBF	1	<6>	0.503	0.000	0.503	SXX_SP - Fab. Spool	
S03_SBF	1	<5>	0.089	0.000	0.089	SXX_SP - Fab. Spool	

NOTES :

- ALL COORDINATES AND ELEVATIONS ARE IN 'METER' AND ALL DIMENSIONS ARE IN 'MM'.
- PIPE QUANTITY IS IN METERS AND OTHER QUANTITY IN NUMBERS.
- FOR STANDARD SUPPORT, REFER DOC. NO. "PP-866-ST-0001"
- IN CASE OF ANY DISCREPANCY FOR LINE CONDITIONS, LINE DATA GIVEN IN LINE LIST SHALL GOVERN
- FOR SMALL BORE LINES, DIMENSION & ROUTING SHALL BE FIELD ADJUSTED.
- THE SPECIFIED THEORETICAL LOCATION OF EQUIPMENT NOZZLES, SUPPORTING STRUCTURE SHALL BE VERIFIED BEFORE INSTALLING THE PIPING & INTOLERABLE DISCREPANCIES SHALL BE DISCUSSED WITH DESIGN SITE ENGINEER.
- ALL THREADED CONNECTION SHALL BE SEAL WELDED, EXCEPT DRAIN AND VENT PLUG.
- REFER LINE LIST.
- ALL PRESSURE WELDS, OTHER THAN BUTT WELDS, INCLUDING BRANCH CONNECTIONS SUCH AS BRANCH-OLETS, AND BOSS) SHALL BE 100% PTMT EXAMINED. ALL ATTACHMENT WELDS ON PRESSURIZED COMPONENTS THAT ARE NOT HYDROTESTED AFTER MAKING THE ATTACHMENT WELD SHALL BE 100% PTMT EXAMINED.

BUTT WELD	FIELD WELD
☒ INSULATED PIPE	☒ TRACED PIPE
☒ PIPE SUPPORT	☒ PLATFORM PENETRATION
☒ SOCKET WELD	☒ SCREWED JOINT

STANDARD DETAIL REFERENCE DRAWINGS

DETAILS HEAVY WELDING ROSSSES FOR THREADED CONNECTIONS	AE-036175
CONSTRUCTION DETAILS AND DETAILS FOR ISOMETRIC PIPING	AD-036177
INSULATION DETAILS FOR HOT PIPING	AA-036181
INSULATION DETAILS FOR COLD PIPING	AA-036183
INSTALLATION OF JACK SCREWS FOR FLANGED JOINTS	AC-036180

NOTES

- MATERIAL SHOWN WITH A BROKEN LINE AND SPOOLS LISTED IN THE SAUDI ARAMCO FABRICATION LIST ARE NOT TO BE FURNISHED BY THE VENDOR.
- PIPING DETAILS SHALL BE CUT AT ALL POINTS MARKED F.W. (FIELD WELD).
- EACH FABRICATION SECTION SHALL BE CLEARLY MARKED WITH ITS CODE NUMBER.
- UNLESS OTHERWISE NOTED: (A) ALL DIMENSIONS ARE TO FACE OF FLANGES. (B) BOLT HOLES ON FLANGES STRADDLE C. (C) WELDING ELBOWS ARE LONG RADIUS.
- FABRICATION SHALL BE IN ACCORDANCE WITH SAUDI ARAMCO SPECIFICATION 01-SAMSS-10.
- GASKET ALLOWANCE
- PIPING MATERIAL IS

PIPE SIZE	SCHEDULE	SPECIFICATION

9. WELDING STRESS RELIEF - YES ☐ NO ☐ 100% X-RAY ☐ SPOT X-RAY 01-SAMSS-10 ☐
9. TESTING SHOP HYDROSTATIC TEST - YES ☐ NO ☐ THE SHOP HYDROSTATIC TEST SHALL BE CARRIED OUT ON PIECES HAVING ALL ENDS FLANGED. HYDROSTATIC TEST PRESSURE: _____
10. ALL PIPING MATERIAL IS IN ACCORDANCE WITH SAUDI ARAMCO ENGINEERING SPECIFICATION SAES-L-105 CLASS: _____ ALL EXCEPTIONS ARE NOTED ON THE ISOMETRIC.
11. INSULATION REQUIRED - YES ☐ NO ☐ INSULATION SHOWN ON DRAWING: _____

SPOOLS TO BE FABRICATED BY VENDOR

NO. OF PIECES	MARK NO.	SIZE	WEIGHT LBS	NO. OF PIECES	MARK NO.	SIZE	WEIGHT LBS

SPOOLS TO BE FABRICATED BY SAUDI ARAMCO

NO. OF PIECES	MARK NO.	SIZE	NO. OF PIECES	MARK NO.	SIZE

REFERENCE DRAWINGS

PIPING PLAN	
P&ID NO.	YA-221234-001
PLOT PLAN KEY PLAN NO.	TA-293108
INDEX FOR PIPING ISO (AG)	PP-865-LI-0003
INDEX FOR PIPING ISO (UG)	PP-865-LI-0004

REV	DATE	DESCRIPTION	PROD BY	CHKD BY	APP BY	APP BY
00	29-09-25	ISSUED FOR CONSTRUCTION	AC/SN	KY/PD	BSP	HJ

REVISION STATUS

REVISION VALIDATION	
THIS REVISION NO 00 IS COVERED, FOR ALL APPROVAL AND CERTIFICATION REQUIREMENTS, PER SAEP-33A, UNDER DRAWING COMPLETION CERTIFICATE NO: 10-14815-0003-000	

DESIGN CERTIFICATION		REVIEW FOR KEY DRAWINGS		OTHER	
ENGG./DATE	CERTIFIED BY/DATE	CONSTR. AGENCY/DATE	OPERATIONS/DATE	BY/DATE	
REV. NO.	DATE	REVISION DESCRIPTION			

JOB ORDER/REV. NO. 10-14815-0003

THIS REVISION IS NOT TO BE USED FOR CONSTRUCTION UNLESS CERTIFIED AND DATED

SAUDI ARABIAN OIL COMPANY

ISOM 1" - IA-00970-1CG0U01

IA SERVICE

JFGP GAS PROCESSING TRAIN-2

JAFURAH SAUDI ARABIA

PLANT NO. 865 INDEX L DRAWING NUMBER YB-265230 SHT. NO. 010f02 REV. NO. 00

SITE REV. NO. 001

RESTRICTED

SMART

CADD-03